

# Week 4: Counting in Base 2

JSS1 Mathematics — First Term Lesson Note (Deep-Dive Edition)

## Introduction

Every number system has a "base" — the number of unique digits it uses. We use **base 10** in everyday life because we have ten fingers, but computers use **base 2 (binary)** because electronic circuits only have two possible states: ON and OFF.

## What is a Number Base?

The number base tells us how many digits are used before we "carry over" to a new column. Some common bases include: base 2 (binary), base 8 (octal), base 10 (denary/decimal), and base 16 (hexadecimal) — but in JSS1 we focus on **base 2**, which uses only **two digits: 0 and 1**.

## Counting in Binary

| Denary (Base 10) | Binary (Base 2) |
|------------------|-----------------|
| 0                | 0               |
| 1                | 1               |
| 2                | 10              |
| 3                | 11              |
| 4                | 100             |
| 5                | 101             |
| 6                | 110             |
| 7                | 111             |
| 8                | 1000            |
| 9                | 1001            |
| 10               | 1010            |
| 11               | 1011            |
| 12               | 1100            |
| 13               | 1101            |
| 14               | 1110            |
| 15               | 1111            |
| 16               | 10000           |

## Place Value in Base 2

In binary, each column is a power of 2 (from right to left): ...32, 16, 8, 4, 2, 1

**Example 1:** The binary number 1011 means:

$$(1 \times 8) + (0 \times 4) + (1 \times 2) + (1 \times 1) = 8 + 0 + 2 + 1 = \mathbf{11 \text{ in base 10}}$$

**Example 2:** The binary number 1101 means:

$$(1 \times 8) + (1 \times 4) + (0 \times 2) + (1 \times 1) = 8 + 4 + 0 + 1 = \mathbf{13 \text{ in base 10}}$$

## Real-Life Application

Every photo, song, and message on a phone or computer is ultimately stored as a long sequence of 1s and 0s (binary digits, or "bits"), because computer memory is built from millions of tiny switches that are either ON (1) or OFF (0).

## Vocabulary

| Word   | Meaning                                                          |
|--------|------------------------------------------------------------------|
| Bit    | A single binary digit — either 0 or 1 (short for "binary digit") |
| Base   | The number of unique digits used in a number system              |
| Denary | Another name for base 10 (the everyday counting system)          |

## Evaluation Questions

- 1 Count from 0 to 20 in binary.
- 2 Write the binary place values for the number 10110.
- 3 Why do computers use base 2 instead of base 10?
- 4 Convert the binary number 1110 to base 10.
- 5 What is a "bit," and why is it important in computing?
- 6 Name two other number bases apart from base 2 and base 10.
- 7 Convert the binary number 10011 to base 10.
- 8 Explain, in your own words, what "place value in base 2" means.
- 9 Why does binary only need two digits, while denary needs ten?
- 10 Write the base 10 number 6 as a binary number, using the counting table above.